

Obesity Paradox in the ICU: A Causal Analysis of Effect Heterogeneity

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Course

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Abstract

This report presents a causal analysis of the “obesity paradox” in critical care using MIMIC-IV electronic health record data ($N = 21,766$ ICU admissions). Employing increasingly robust estimators from naive association through doubly-robust cross-fitted AIPW (DML), the report estimates the causal effect of elevated BMI (overweight or obese: $\text{BMI} \geq 25$) on in-hospital mortality in the ICU. The aggregate average treatment effect is small, negative, and statistically non-significant once confounding is addressed (DML: $\text{ATE} = -0.004$, $p = 0.24$). However, the analysis reveals statistically significant heterogeneity by sex ($p = 0.009$): the protective association is concentrated among female patients ($\text{z-TE} = -0.014$, $p = 0.009$) and absent among males ($\text{z-TE} = +0.004$, $p = 0.35$). No evidence of effect modification by admission type (medical vs. surgical) was found. Sensitivity analyses using E-values and an alternative BMI threshold ($\text{BMI} \geq 30$) confirm the robustness of the primary sex-stratified finding.

1 Data

This analysis uses electronic health record data from MIMIC-IV, a large publicly available critical care database, accessed through the `ricu` R package. MIMIC-IV contains de-identified records from adult patients admitted to the Beth Israel Deaconess Medical Center ICU in Boston, MA, from 2008 to 2019. The study population is restricted to adult patients with their first ICU admission and valid BMI values between 18.5 and 35 kg/m², excluding extreme underweight and morbid obesity to ensure comparability. The final analytic cohort comprises $N = 21,766$ patients: 18,747 medical admissions and 3,019 surgical admissions.

Key variables are as follows:

- **Treatment (X):** Elevated BMI indicator (overweight or obese): $\text{BMI} \geq 25$ (prevalence 60.4%)
- **Outcome (Y):** In-hospital mortality (binary; overall rate $\approx 7.0\%$)
- **Adjustment covariates (Z):** Age, Sex, Charlson comorbidity proxy (ICD-code count)
- **Effect modifiers:** Sex (primary) and admission type — medical vs. surgical (secondary)

Cohort summary: 9,350 female and 12,416 male patients; propensity scores ranged from 0.515 to 0.688 with 0% of values outside (0.05, 0.95), confirming strong covariate overlap.

Table 1: Cohort Characteristics

Variable	Value
Sample size (N)	21,766
Medical / Surgical	18,747 / 3,019
Overall mortality	7.0%
Overweight/obese ($\text{BMI} \geq 25$)	60.4%
Female / Male	9,350 / 12,416
PS range	[0.515, 0.688]

2 Causal Query

We study the “obesity paradox” in critical care: the counterintuitive finding that overweight or obese patients ($\text{BMI} \geq 25$) may have lower ICU mortality than normal-weight counterparts. The primary causal estimands are defined as risk differences under the potential outcomes framework:

- **ATE** (Average Treatment Effect): $\mathbb{E}[Y(1) - Y(0)]$ — effect averaged over the full cohort
- **z-TE** (Subgroup Effects): $\mathbb{E}[Y(1) - Y(0) \mid Z = z]$ — effect within levels of sex and admission type

The central research question is: *Does the causal effect of elevated BMI on ICU mortality vary across subpopulations, primarily by sex, and secondarily by admission type (medical vs. surgical)?* Prior literature on the obesity paradox has been criticized for pooling heterogeneous subgroups; this analysis directly targets that heterogeneity.

3 Causal Model

3.1 Directed Acyclic Graph (DAG)

The assumed causal structure is encoded in the DAG below. Age, sex, and comorbidities (Charlson proxy) are common causes of both BMI category and mortality, making them confounders. Admission type is treated as a potential effect modifier but, consistent with MIMIC-IV’s predominantly emergency cohort, is assumed not to lie on a causal pathway from BMI to mortality. Unmeasured confounders (U), such as illness severity scores (SOFA) and medications, are acknowledged as unobserved in the study.

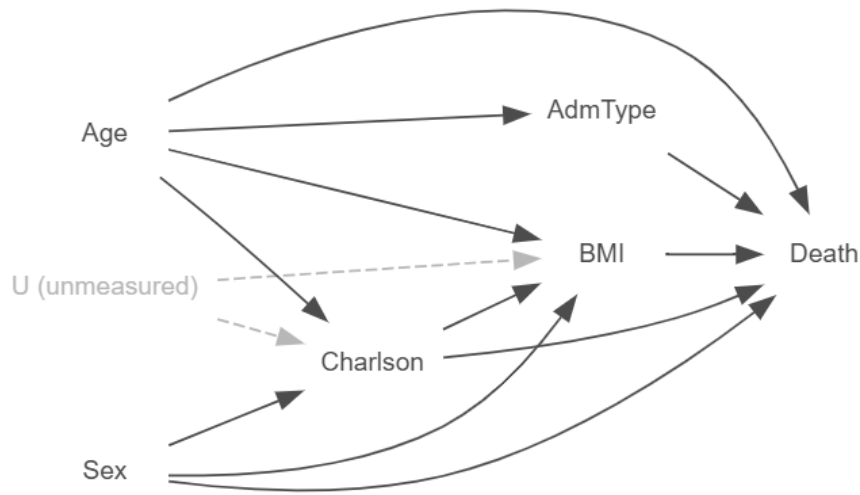


Figure 1: DAG for the elevated BMI–mortality relationship. Arrows denote direct causal effects. U = unmeasured confounders (e.g., SOFA score, medications).

DAG Justification

Each edge in the DAG is motivated by clinical and epidemiological reasoning:

- **Age → BMI:** Metabolic changes with aging; older patients are more likely to fall in the overweight or obese BMI range.
- **Age → Mortality:** Age is one of the strongest independent predictors of in-hospital mortality in critically ill patients.
- **Sex → BMI:** Biological sex influences fat distribution and BMI through hormonal and physiological mechanisms, producing different BMI distributions across male and female populations.
- **Sex → Mortality:** Different inflammatory and immune responses between male and female patients, especially in the ICU.

- **Charlson $\rightarrow$ BMI:** Comorbidity burden can causally affect body weight; chronic conditions such as heart failure, diabetes, and other diseases are associated with both weight gain or disease-related weight loss, linking comorbidity to BMI.
- **Charlson $\rightarrow$ Mortality:** Higher comorbidity burden directly increases mortality risk independent of BMI.
- **BMI $\rightarrow$ Mortality:** The causal pathway of interest. Elevated BMI may confer protective metabolic reserves during critical illness, motivating the obesity paradox hypothesis.
- **Age, Charlson $\rightarrow$ Admission type:** Older and sicker patients are more likely to present via medical pathways; surgical admissions tend to be younger and have lower baseline comorbidity burden.
- **$U \rightarrow$ BMI, $U \rightarrow$ Mortality:** Unmeasured confounders, most notably illness severity (SOFA score), frailty, and disease-related weight loss prior to admission, are plausible common causes of both observed BMI and mortality. SOFA is treated as post-baseline and excluded from the primary adjustment set to avoid conditioning on a potential mediator.

Edges that are *absent* from the DAG are equally important. Admission type is assumed not to cause BMI in this cohort: ICU admission type is determined by the presenting clinical condition, not by the patient’s body weight. This assumption is most defensible in the predominantly emergency setting of MIMIC-IV iCU data, though it remains a contestable structural choice in the analysis.

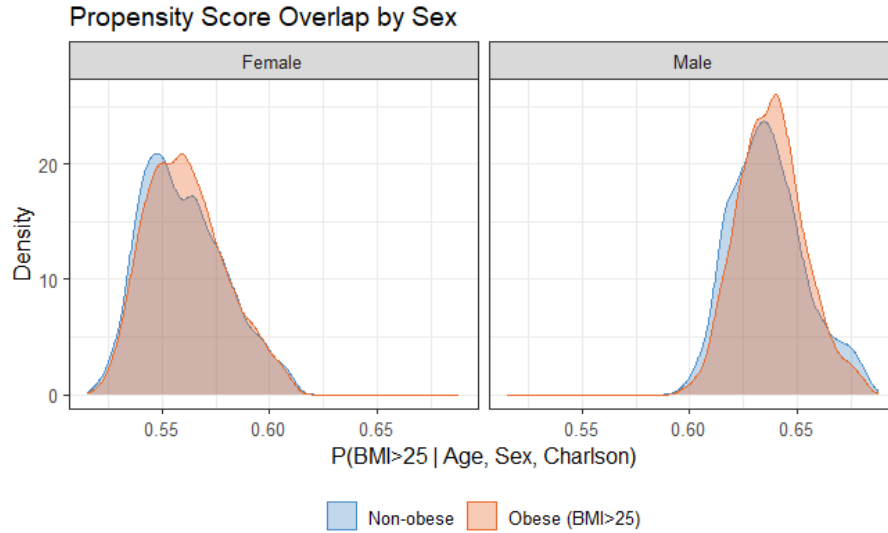


Figure 2: Propensity score overlap by sex. Estimated $\hat{e}(Z) = P(X=1 \mid Z)$ from a logistic model in $\{\text{Age, Sex, Charlson}\}$. All scores fall within $[0.515, 0.688]$, confirming the positivity assumption.

3.2 Identification Assumptions

Identification of the ATE via the backdoor criterion requires three assumptions:

- (i) **Consistency:** $Y = Y(x)$ when $X = x$ — the observed outcome equals the potential outcome under the treatment received.
- (ii) **Backdoor criterion satisfied:** The adjustment set $Z = \{\text{Age, Sex, Charlson}\}$ satisfies Pearl’s backdoor criterion in the assumed DAG; it blocks all backdoor paths from X to Y without conditioning on any descendant of X , licensing the adjustment formula $P(Y \mid do(X)) = \sum_z P(Y \mid X, Z=z) P(Z=z)$. Violated if U opens an unblocked backdoor path.
- (iii) **Positivity:** $P(X = x \mid Z = z) \in (0, 1)$ for all z . Empirically satisfied: PS $\in [0.515, 0.688]$, no extremes.

By the backdoor criterion, conditioning on $Z = \{\text{Age, Sex, Charlson}\}$ blocks all open backdoor paths from X to Y . Admission type is not a confounder under the primary DAG and is therefore excluded from the causal adjustment set Z ; it is however included as a baseline covariate in the nuisance models (propensity score and outcome models) to improve estimation precision, and serves as the stratification variable for the secondary subgroup analysis.

4 Identification, Estimation, and Sensitivity Analysis

4.1 Estimation Methods

We employ an estimation ladder of five progressively more robust estimators, improving over the naive association by introducing covariate adjustment, doubly-robust correction, and cross-fitting to reduce overfitting bias. Agreement across estimators strengthens causal credibility.

Table 2: Estimation Methods Summary

Estimator	Method	Key Property
TV (Naive)	Raw mortality difference	No confounding adjustment; baseline only
IPW	Inverse probability weighting (logistic PS)	Adjusts via reweighting; sensitive to PS misspecification
OR (GLM)	Outcome regression (logistic $Y \sim X + Z$)	Adjusts via outcome model; sensitive to outcome misspecification
AIPW	Doubly-robust combination of IPW + XGB outcome model	Consistent if either PS or outcome model is correct
DML (CF-AIPW)	Cross-fitted AIPW; nuisance via XGB (5-fold)	Reduces overfitting bias; primary inferential estimator

The DML estimator is based on the AIPW pseudo-outcome for each unit i :

$$\varphi_i = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1 - X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)}$$

where $\hat{e}(Z_i) = \hat{P}(X = 1 \mid Z_i)$ is the estimated propensity score, $\hat{\mu}_1(Z_i)$ and $\hat{\mu}_0(Z_i)$ are the estimated potential outcome means under treatment and control, and the ATE is

then $\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \varphi_i$. Cross-fitting ensures that nuisance estimates $\hat{e}$ and $\hat{\mu}$ are trained on held-out folds, preventing overfitting bias.

For effect heterogeneity, we use stratified CF-AIPW (z-TE estimates) with Wald tests to assess whether the treatment effect differs across subgroups. As an exploratory addition, we fit a causal forest (2) to estimate individual-level CATEs nonparametrically and test for the presence of heterogeneity using the best linear projection test.

4.2 Sensitivity Analysis

Two robustness checks address key threats to validity:

- **E-values** (1): Quantify the minimum strength of an unmeasured confounder (in terms of risk ratio) that would be required to fully explain away the estimated effect. Higher E-values indicate greater robustness.
- **Treatment threshold sensitivity:** We re-run all analyses using $\text{BMI} \geq 30$ (the clinical threshold for obesity) rather than $\text{BMI} \geq 25$ (overweight or obese) to assess whether results depend on the chosen cut-off point.

5 Results

5.1 Overall ATE: The Obesity Paradox is Weak in Aggregate

Across all estimators, the ATE is small and negative (protective direction). In less robust estimation methods, there is a statistically significant causal effect detected, pointing toward obesity lowering mortality rate. Under the doubly robust estimation methods (AIPW and DML), a small negative ATE is detected, but statistically not significant.

Table 3: ATE Estimates Across the Estimation Ladder (Risk Differences)

Estimator	ATE	95% CI
TV (Naive)	−0.010	[−0.018, −0.006]
IPW	−0.006	[−0.014, −0.002]
OR (GLM)	−0.005	[−0.014, −0.002]
AIPW	−0.004	[−0.012, +0.004]
DML (CF-AIPW)	−0.004	[−0.011, +0.003]

The consistent attenuation from TV (−0.010) to DML (−0.004) reflects confounding adjustment: once age, sex, and comorbidities are controlled for, the apparent protective effect shrinks substantially. The DML estimate is not significant ($p = 0.24$), indicating the overall obesity paradox is weak and statistically uncertain in this cohort.

5.2 Effect Heterogeneity by Sex: Primary Finding

The central finding of this analysis is a statistically significant difference in the effect of elevated BMI by sex ($p = 0.009$):

Table 4: z-TE by Sex (CF-AIPW). Negative values indicate elevated BMI reduces mortality risk.

Subgroup	z-TE Estimate	SE	95% CI	p-value
Female	−0.0141	0.0054	[−0.025, −0.004]	0.009
Male	+0.0043	0.0045	[−0.005, +0.013]	0.349
Difference (F − M)	−0.0183	0.0070	[−0.032, −0.005]	0.009

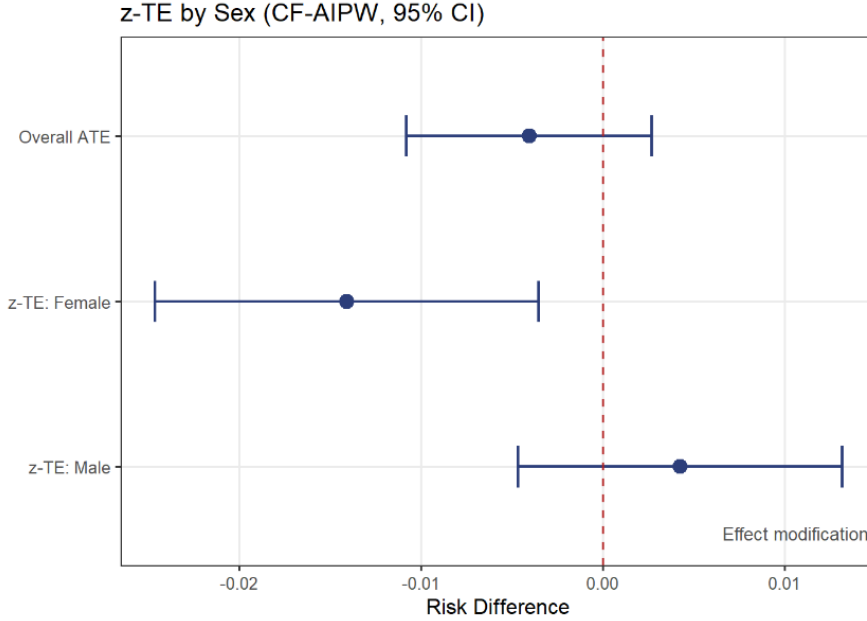


Figure 3: CF-AIPW z-TE estimates by sex with 95% confidence intervals. Female patients show a significant protective effect (−1.4 pp, $p = 0.009$); males show a near-zero, non-significant effect. Dashed line at zero; effect modification $p = 0.009$.

Among female ICU patients, elevated BMI is associated with a significant reduction in mortality risk (−1.4 percentage points). Among male patients, the point estimate is near zero and non-significant. The sex difference of −1.83 pp is robustly significant ($p = 0.009$). Even after a Bonferroni correction for multiple subgroup tests ($\alpha_{adj} = 0.025$), this heterogeneity remains statistically significant ($p_{adj} = 0.018$). This pattern is consistent with proposed biological mechanisms: estrogen-mediated anti-inflammatory effects, differences in fat distribution (subcutaneous vs. visceral adiposity), and sex-specific immune responses during critical illness.

5.3 Effect Heterogeneity by Admission Type: Null Secondary Finding

No evidence of effect modification by admission type was found ($p = 0.69$). The z-TE estimates are nearly identical for medical (−0.0035) and surgical (−0.0011) patients, and the confidence intervals overlap substantially. This null finding is itself informative: the obesity paradox in this cohort is not driven by differences between medical and surgical pathways.

5.4 Causal Forest: Exploratory Heterogeneity

The causal forest (mean CATE = -0.0027) provides converging evidence of heterogeneity: the best linear projection test yields a significant differential prediction coefficient ($p = 0.042$). The best linear projection further confirms sex as the primary driver of CATE variation ($p = 0.009$), with age, Charlson score, and admission type non-significant. The CATE-by-decile plot shows CATEs ranging from approximately -0.06 (most protective) to $+0.06$ (potentially harmful), consistent with the presence of meaningful subpopulation variation.

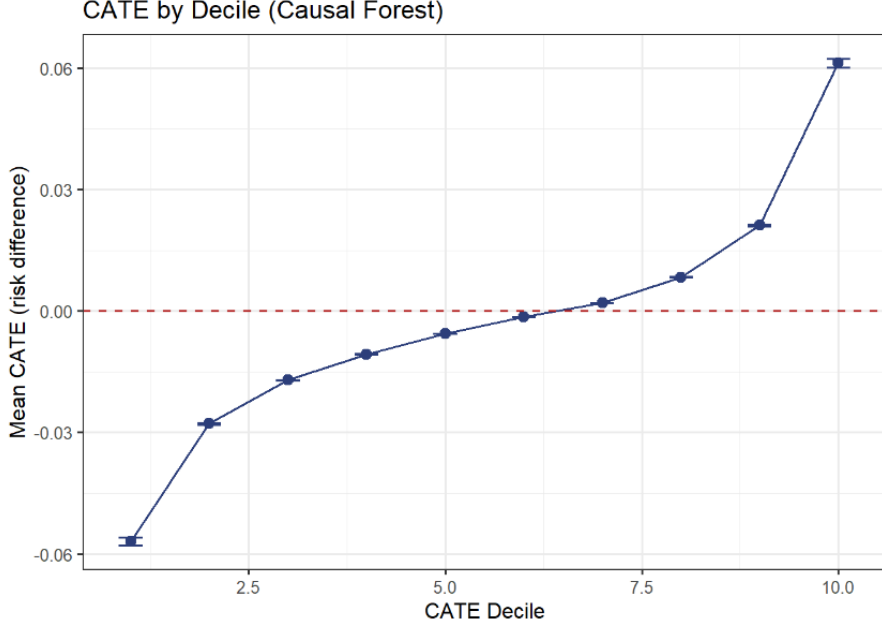


Figure 4: Mean CATE by decile from the causal forest (2,000 trees, seed 42). Error bars are $\pm 1.96 \cdot \text{sd}(\hat{\tau}_i) / \sqrt{n_{\text{dec}}}$. CATEs range from approximately -0.06 (most protective) to $+0.06$ (potentially harmful), consistent with meaningful individual-level heterogeneity.

5.5 Sensitivity Analysis

E-values confirm robustness to unmeasured confounding:

Table 5: E-Values Quantifying Robustness to Unmeasured Confounding

Estimand	RD Estimate	E-value	E-value (CI bound)
Overall ATE	-0.0040	1.30	1.23
z-TE: Female	-0.0141	1.70	1.26
z-TE: Medical	-0.0035	1.25	1.28

Male z-TE omitted: CI crosses null; no confounding needed to include null.

The E-value for the female z-TE is 1.70, meaning an unmeasured confounder would need to be associated with both elevated BMI and mortality by a risk ratio of at least 1.70 on both sides to fully explain away the protective effect. This represents a moderate degree of robustness given the modest effect sizes.

Under the $\text{BMI} \geq 30$ threshold: $\text{ATE} = -0.0065$, $\text{z-TE}(\text{female}) = -0.0158$, $\text{z-TE}(\text{male}) = +0.0018$, sex difference $p = 0.010$. The pattern of female-specific protection and the significance of sex-based heterogeneity persists under the stricter clinical definition of obesity, supporting the robustness of the primary finding.

6 Conclusions and Limitations

6.1 Conclusions

This study contributes a causal analysis of the obesity paradox in critical care. Three main conclusions emerge:

- (i) The overall causal effect of elevated BMI ($\text{BMI} \geq 25$) on ICU mortality is small, negative, and statistically non-significant once confounding is addressed via doubly-robust methods.
- (ii) There is strong, statistically significant heterogeneity by sex: the protective association is concentrated in female patients ($\text{z-TE} = -0.014$, $p = 0.009$) and is absent in males ($\text{z-TE} = +0.004$, $p = 0.35$). This finding is robust to the BMI threshold definition ($\text{BMI} \geq 30$: sex difference $p = 0.010$).
- (iii) No evidence of effect modification by admission type was found, suggesting the paradox is not driven by differences between medical and surgical patient admissions to the ICU.

These findings suggest that prior aggregate estimates of the obesity paradox may mask opposing effects across subgroups. Clinically, this motivates sex-stratified investigation of the mechanisms underlying elevated-BMI-related resilience in critical illness.

6.2 Limitations

- **Charlson proxy:** `ricu` returns ICD-code counts as a comorbidity proxy rather than a validated Charlson index, potentially attenuating adjustment quality for comorbidity burden.
- **BMI measurement timing:** BMI is sourced from outpatient records (OMR table), not ICU-day weight, introducing possible misclassification — particularly for fluid-overloaded patients.
- **Missing severity scores:** SOFA scores were unavailable in this MIMIC-IV extraction, precluding adjustment for illness severity — a potentially important unmeasured confounder.
- **Unmeasured confounding:** E-values indicate only moderate robustness (≈ 1.3 – 1.7), and the exchangeability assumption cannot be fully verified.
- **Causal forest standard errors:** CATE standard errors from $\text{sd}(\text{CATE})/\sqrt{n}$ understate true uncertainty; the AIPW-based z-TE estimates in Sections 5.2–5.3 are the valid inferential results for effect heterogeneity.

References

- [1] VanderWeele, T. J. and Ding, P. (2017), ‘Sensitivity analysis in observational research: Introducing the E-value’, *Annals of Internal Medicine* **167**(4), 268–274.
- [2] Wager, S. and Athey, S. (2018), ‘Estimation and inference of heterogeneous treatment effects using random forests’, *Journal of the American Statistical Association* **113**(523), 1228–1242.

Code Availability

Supporting code and implementation details are available at:

<https://github.com/ajyamin/c160finalproject>.